

Level Mudah

Tugas 1 – Membuat Queue Sederhana Menggunakan List

```
queue = []

queue.append("Andi")
queue.append("Budi")
queue.append("Citra")

print("Isi Queue:")
print(queue)
```

```
Isi Queue:
['Andi', 'Budi', 'Citra']
```

Tugas 2 – Implementasi Enqueue dan Dequeue

```
queue = []

# Enqueue
queue.append("Andi")
queue.append("Budi")
queue.append("Citra")

print("Queue Awal:", queue)

# Dequeue
layani = queue.pop(0)

print("Yang Dilayani:", layani)
print("Queue Sekarang:", queue)
```

```
Queue Awal: ['Andi', 'Budi', 'Citra']
Yang Dilayani: Andi
Queue Sekarang: ['Budi', 'Citra']
```

Tugas 3 – Menampilkan Front dan Rear

```
queue = ["Andi", "Budi", "Citra",]

print("Queue:", queue)
print("Front:", queue[0])
print("Rear :", queue[-1])
```

```
Queue: ['Andi', 'Budi', 'Citra']
Front: Andi
Rear : Citra
```

Level Menengah

Tugas 4 – Menu Queue Interaktif

```
queue = []

while True:
    print("\n=== MENU QUEUE ===")
    print("1. Tambah Antrian")
    print("2. Layani Antrian")
    print("3. Lihat Antrian")
    print("4. Keluar")

    pilihan = input("Pilih Menu: ")

    if pilihan == "1":
        nama = input("Masukkan Nama: ")
        queue.append(nama)
        print(nama, "masuk antrian")

    elif pilihan == "2":
        if len(queue) == 0:
            print("Antrian kosong")
        else:
            print("Dilayani:", queue.pop(0))
```

```

elif pilihan == "3":
    print("Daftar Antrian:", queue)

elif pilihan == "4":
    print("Program selesai")
    break

else:
    print("Pilihan tidak valid")

```

```

=== MENU QUEUE ===
1. Tambah Antrian
2. Layani Antrian
3. Lihat Antrian
4. Keluar
Pilih Menu: 1
Masukkan Nama: Musa
Musa masuk antrian

```

```

=== MENU QUEUE ===
1. Tambah Antrian
2. Layani Antrian
3. Lihat Antrian
4. Keluar
Pilih Menu: 2
Dilayani: Musa

```

```

=== MENU QUEUE ===
1. Tambah Antrian
2. Layani Antrian
3. Lihat Antrian
4. Keluar
Pilih Menu: 3
Daftar Antrian: []

```

```

=== MENU QUEUE ===
1. Tambah Antrian
2. Layani Antrian
3. Lihat Antrian
4. Keluar
Pilih Menu: 4
Program selesai

```

Tugas 5 – Simulasi Antrian Kasir Minimarket

```

queue = []

queue.append("Pelanggan 1")
queue.append("Pelanggan 2")
queue.append("Pelanggan 3")
queue.append("Pelanggan 4")

print("Antrian Awal:")
print(queue)

print("\nKasir melayani...")
print("Dilayani:", queue.pop(0))

print("\nSisa Antrian:")
print(queue)

```

```

Antrian Awal:
['Pelanggan 1', 'Pelanggan 2', 'Pelanggan 3', 'Pelanggan 4']

```

```

Kasir melayani...
Dilayani: Pelanggan 1

```

```

Sisa Antrian:
['Pelanggan 2', 'Pelanggan 3', 'Pelanggan 4']

```

Tugas 6 – Implementasi Queue Menggunakan deque

```

from collections import deque

queue = deque()

# Enqueue
queue.append("Andi")
queue.append("Budi")
queue.append("Citra")

print("Queue Awal:")

```

```
print(queue)

# Dequeue
layani = queue.popleft()

print("\nYang Dilayani:", layani)
print("Queue Sekarang:")
print(queue)
```

Queue Awal:
deque(['Andi', 'Budi', 'Citra'])

Yang Dilayani: Andi
Queue Sekarang:
deque(['Budi', 'Citra'])

Bonus (Tugas 7) – Perbandingan Queue List dan Deque

```
from collections import deque

# Queue menggunakan List
queue_list = ["A", "B", "C"]
queue_list.pop(0)

# Queue menggunakan Deque
queue_deque = deque(["A", "B", "C"])
queue_deque.popleft()

print("List :", queue_list)
print("Deque:", queue_deque)
```

List : ['B', 'C']
Deque: deque(['B', 'C'])